TEMPORALLY CONSISTENT SOCCER FIELD REGISTRATION

Antje Linnemann, Sebastian Gerke, Sven Kriener, Patrick Ndjiki-Nya

Image Processing Department
Fraunhofer Institute for Telecommunications, Heinrich Hertz Institute
{antje.linnemann, sebastian.gerke, patrick.ndjiki-nya}@hhi.fraunhofer.de

ABSTRACT

In this paper, a system for temporally consistent registration of a soccer field is proposed. It is implemented using 2D field line and arc detection as well as motion estimation for frames where too few field lines are detected. We show that a combination of these technologies yields more consistent results than common state-of-the-art approaches. As an important additional contribution, a Kalman filter is applied to suppress remaining misdetections. The system is able to register the soccer field with a mean error of 2.53 m regardless of the visible field region.

***Index Terms*—** Soccer video analysis, play field registration

1. INTRODUCTION

With today's progress in communication technology, broadcasters have to deal with an increasing quantity of live and archived sports video material. Therefore, tools for the automatic analysis of this type of content are strongly required. Especially, technologies for automatic soccer analysis are of high interest since this sport attracts millions of people worldwide. For instance, player localisation and recognition as well as highlight detection are essential for effective search and summarization of soccer content. To obtain such semantic information, the homography from the image onto the real-world playfield plane is highly beneficial. It enables the computation of real-world positions of players and ball. Furthermore, prior knowlegde on where the camera is focusing proved to be a useful feature for highlight detection [1, 2, 3, 4]. For both applications temporally consistent results are more important than the absolute accuracy per frame.

Today, several approaches exist for computation of the homograpy between the image plane and a given playfield model. Most of them use intersections of field lines as point correspondences. For instance, Farin et al. [5] proposed an automatic camera calibration algorithm for court sports based on the detection of straight lines. However, this technology is inadequate for registration of soccer play fields, as intersections of straight lines cannot be extracted from most images. If insufficient point correspondences have been found, a common approach is to use global motion to estimate the homography from neighbouring frames [2, 6]. For the same reason, arc detection is often used in addition to line detection. This enables reliable field registration for center regions, even if just the center circle and line are visible [7, 8]. In fact, Wang et al. rely entirely on the detection of arcs, as their system detects the arc of the penalty box as well [9].

In this paper, a system for a reliable, temporally consistent registration of a soccer play field is proposed. It is capable to compute the homography regardless of the view angle. In contrast to existing approaches, it is the first that ensures temporal consistency by employing a Kalman Filter on homographies obtained from circle and arc detection.

2. PROPOSED TECHNOLOGY

The goal of soccer field registration is to find a perspective homograpy $\mathbf{H}$, which transforms the real-world coordinates $\mathbf{p}_w$ to image coordinates as $\mathbf{p}_i = \mathbf{H}\mathbf{p}_w$. The used homography is perspective and thus non-linear and possesses eight parameters, which pairwise describe translation, scaling, shearing and perspective distortions. To determine the transformation matrix H, at least four image points of which real-world coordinates are known, have to be found. Our system relies on the play field model given in [10] in order to use the intersections of field lines as candidates. We restrict our soccer field registration approach to overview scenes, as required line features are usually unavailable in close-ups or intermediate scenes.

(a) goal view (b) center view (c) intermediate

Fig. 1. Possible view angle types

But, in most frames only a part of the soccer field is vis-

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ible. As depicted in figure 1, an overview frame typically shows either the goal region with straight lines in the penalty box, the center region with a center line and a center circle, or an intermediate region with no usable line intersections at all. Due to their diverse characteristics, it is appropriate to design dedicated tools for each type of view angle.

Hence, the main contribution of our system is the view angle optimized processing, which is achieved by distinguishing three view angle classes. Furthermore, a Kalman filter is applied to suppress the influence of remaining misdetection.

Figure 2 depicts the proposed system, with bold blocks and arrows emphasizing the main contributions of this paper. After field line detection, each image is assigned to one of the following classes: goal, center or intermediate view. Depending on class assignment, the system determines the homography with the most promising estimation approach. For goal and center view, the homography is computed directly from the image. This is an iterative process which identifies the best valid homography. If no homography can be found or an intermediate view is identified, the homography is determined through global motion estimation. Finally, a Kalman filter is applied to all results.

The individual steps will be described more precisely in the following sections.

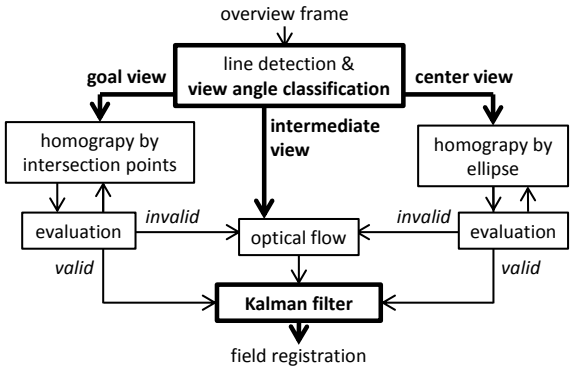


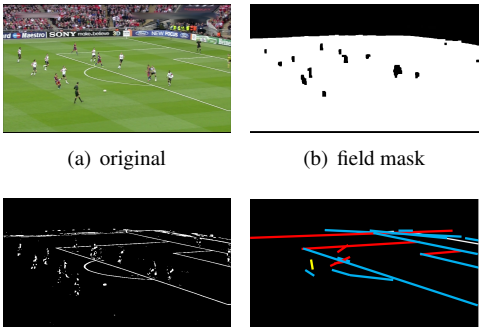
Fig. 2. Block diagram of the field registration approach (bold blocks and arrows are main contributions)

2.1. Line detection

The system depends on a reliable detection of field lines to obtain suitable point correspondences.

First, a playfield mask is generated by adaptive thresholding pixel values depending on the most prominent color in the H channel of the HSV color space representation of a frame. Afterwards, a morphological opening and closing along with a floodfill from the image center is performed. An example for the field mask is shown in figure 3(b).

For line detection, the probabilistic Hough transform is performed on a white line mask as given in figure 3(c). For creation of this mask, lines violating specifications for color and width are ignored [5].



(a) original (b) field mask

(c) line mask (d) line detection and fusion

Fig. 3. Field line detection steps

Line segments that are close to each other and whose directions are similar are fused by a weighted mean. Finally, lines are classified as vertical (yellow), slash (red) and backslash (blue), as shown in figure 3(d).

2.2. View angle classification

A rule-based classifier is necesssary to identify which region of the play field is visible in the image. Direction and length of the previously detected lines are used as features. Decisions are made by the rules stated in table 1. Depending on this classification, different homography computation methods are performed.

View class	Rule
Center	one or more vertical lines with length $> l_c$
Goal	no center region and 4 or more slash and backslash lines each with length $> l_g$
Intermediate	all other cases

Table 1. Rule-based view angle classifier

2.3. Direct homography computation

For goal views, the four longest slash and the four longest backslash lines are used to generate candidate intersection points. These intersections can also lie outside the frame or originate from extended field lines. Transformation matrices are calculated for each combination of four points. The determined matrices are validated subsequently. Therefore, it is checked if detected field lines are transformed onto the play field and if the corresponding camera position lies within a plausible range. For each valid transformation matrix, a playfield model is warped onto the image coordinate system (figure 4). Afterwards, it is compared to the line mask shown

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in figure 3(c). The model with the largest white pixels intersection set, relatively to the union of white pixels, is finally selected.

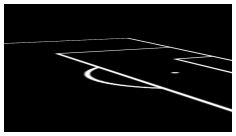

Fig. 4. Field model warped onto the image coordinate system

For center views, it is impossible to find four point correspondences based on straight lines only. However, the center circle and line can be used. According to [7], the center circle can be approximated as an ellipse in the image. Its equation is computed by the algorithm proposed in [11], which requires at least five points on the ellipse to determine its parameters. These points can be extracted through line detection (cf. section 2.1), as tangents of the ellipse are also detected. At first, plausible tangents are located by line length and orientation. Their center points are candidates for the ellipse equation computation. For each combination of points, an ellipse is computed and validated afterwards. Only if the corresponding lines have the expected orientations for tangents and the center of the ellipse is close to the center line, the ellipse equation is further regarded.

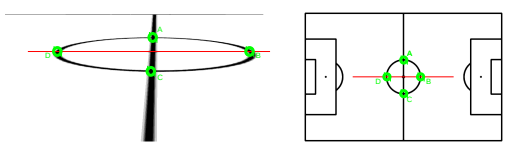

Fig. 5. Point correspondecies on the center circle

For each valid ellipse, the transform matrix is computed by the four points correspondences pictured in figure 5. A transformation matrix is discarded if field lines are not transformed onto the play field or if the corresponding camera position lies outside a plausible range. Finally, the homograpy obtained from the ellipse with the largest number of valid tangents is selected.

2.4. Global motion estimation

When neither the penalty box, nor the center circle is visible, it is impossible to find enough point correspondences to detemine a homography directly. Even, in cases where four points can be found some homograpy computations might yield invalid results. In this situation, the camera motion is estimated by using a Lucas Kanade optical flow estimator on 220 points distributed over the whole image [12]. Dense sampling is used in order to reduce the influence of the players' local motion. For further processing, only points with a valid match are considered. With these points, a motion transformation matrix $\mathbf{M_{n-1,n}}$ is calculated that transforms pixel from frame $n-1$ to frame n. By multiplying this matrix with the perspective transformation matrix $\mathbf{H_{n-1}}$ from the previous frame, an estimation of the current transformation matrix $\mathbf{H_n}$ is obtained:

$$\mathbf{H_n} = \mathbf{M_{n-1,n}}\mathbf{H_{n-1}} \tag{1}$$

Similarly, $\mathbf{M^{-1}_{n,n+1}}$ can be used to to obtain $\mathbf{H_n}$ from the following homography $\mathbf{H_{n+1}}$. Results from both directions are weighted by their distance to the last valid directly computed homography and merged.

3. TEMPORAL CONSISTENCY THROUGH KALMAN FILTERING

Although a view angle dependend homography computation is applied, misdetections may still occur. But, as changes in camera rotation and focus are limited, changes of the visible play field are also restricted. We can thus use this background knowledge to suppress noise in the measurements.

For that, a Kalman filter is proposed to track the visible play field in real world coordinates. This is done by predicting the position of four well-defined points, whose correspondent image coordinates are held constant (e.g. lower right corner of the frame). To reduce complexity, their real-world displacements are assumed to be uncorrelated. In this way, each point can be tracked with a Kalman filter according to [13].

A linear transition model is defined, where a point position (x_t, y_t) is predicted by its previous position (x_{t-1}, y_{t-1}), velocity $(\dot{x}_{t-1}, \dot{y}_{t-1})$ and acceleration $(\ddot{x}_{t-1}, \ddot{y}_{t-1})$.

Position predictions are updated by real measurements. Therefore, the stated image points are transformed to real world coordinates with the directly computed homographies. If no valid homography exists, global motion transformation is applied to the Kalman adjusted homography from the previous frame. The resulting homography is then used to obtain a measurement update.

Finally, the predicted points are used to compute a Kalman adjusted homography.

4. EXPERIMENTAL RESULTS

The proposed technology is applied to a dataset of 28 soccer matches from different leagues. Within this dataset 102 randomly selected frames are used for evaluation. As their annotation is done by manually marking and labeling field line intersection points, annotations are only available for goal and center regions.

The ground truth samples are used to determine the visible area of the pitch for each image. The proposed system is assessed by comparing the automatically computed visible areas to these reference areas. This is done by determining

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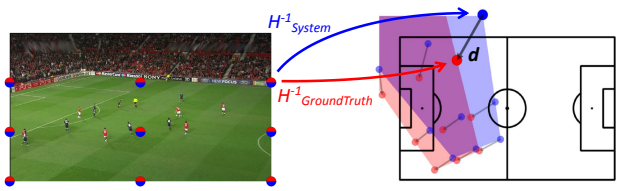

Fig. 6. Example points considered for result evaluation

the euclidean distance between more than 10 000 sampled points and their reference samples as depicted in figure 6. Only points from the lower two-thirds of an image are considered, as upper parts are mostly out of field and are therefore irrelevant for our analysis algorithm.

Table 2 presents the mean disparity between the sampled points and their references for different system configurations. In addition to the mean distance $\bar{d}$ over all sampled frames the first, second and third quartile ($Q_{0.25}, Q_{0.5}, Q_{0.75}$) are also given. All results base upon a play field model of 105 m x 68 m, as given in [10].

Config.	$\bar{d}$	$Q_{0.75}$	$Q_{0.50}$	$Q_{0.25}$
LA	199.07 m	32.62 m	1.43 m	0.64 m
LM	16.27 m	8.45 m	3.88 m	1.14 m
AM	9.70 m	11.54 m	3.85 m	0.77 m
LAM	**3.57m**	**3.83 m**	**1.17 m**	**0.54m**
LAMK	**2.53m**	**2.94 m**	**1.20m**	**0.56 m**

Table 2. Disparity to ground truth

The first two configurations correspond to state-of-the-art approaches. Here, the homography is detemined by intersection points of straight field lines. If the number of point correspondences is insufficient, arc detection is used in configuration *LA*. This is comparable to the system proposed in [7]. Half of the homographies determined by this system configuration have a resulting disparity of less than 1.43 m ($Q_{0.50}$). But the mean error for all homographies is high (199.07 m). This is caused by severe misdetections, which can also be seen in figure 7. Considering that the ground truth does not contain intermediate images, where neither straight lines nor arcs can be detected, we assume that the real mean error should be even higher.

In [2] and [6], motion estimation is used for images where too few straight lines are found. This corresponds to system configuration *LM*. Here, the mean disparity is only 16,27 m, but the median is 3.88 m and thus higher than for configuration *LA*. If arc detection is combined with global motion estimation (configuration *AM*) the median remains almost constant, but the mean error is reduced to 9.70 m. This leads to the assumption that a combination of motion estimation with line and arc detection may yield additive gains.

We thus propose such a combination as given in configuration *LAM*. Our intuition is confirmed by table 3. It shows how often the different approaches are selected for each view angle class. The evaluation step (cf. figure 2) discards about half of the homographies directly determined from goal and center view. By replacing these unreliable detections by global motion estimations, significant improvements are achieved. The *LAM* Configuration yields a mean error of only 3.57 m, with 50% of all homographies resulting in a mean error of less then 1.17 m.

	homography computed by		
ground truth	line det.	arc det.	motion
goal	23	1	28
center	0	26	24

Table 3. Homography computation tool selection for given view classes

To suppress the influence of remaining misdetections, a Kalman filter is applied to the *LAM* system configuration. In this case, the mean error is reduced by 1.04 m to 2.53 m (configuration *LAMK*). The third quartile $Q_{0.75}$ is also reduced by 0.89 m to 2.94 m, whereas the first and second quartile ($Q_{0.25}$ and $Q_{0.50}$) are almost unchanged. So, improvements are mainly achieved for false detections. This is also confirmed by figure 7, as the plots drift further apart with increasing disparity.

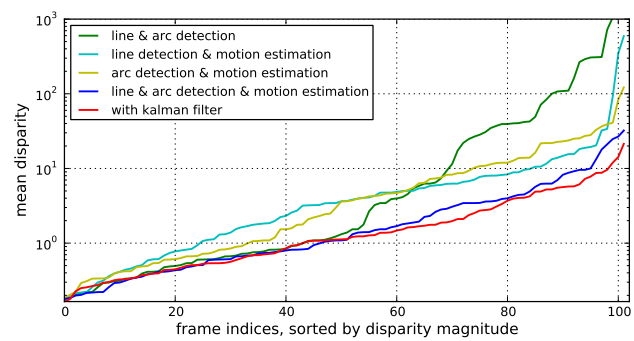

Fig. 7. Log-scaled mean disparity for all annotated frames, sorted by the disparity magnitude (see also configurations *LA*, *LM*, *AM*, *LAM*, *LAMK* in table 2)

5. CONCLUSION

In this paper we showed that registration of a soccer field can be done in a temporally consistent manner for all types of view angles. This was reached by a combination of line and arc detection with global motion estimation. The view dependend processing yielded lower mean disparity than other existing approaches that distinguish less view angle types. Furthermore, we show that applying a Kalman filter to ensure temporal consistency improves the results even further.

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